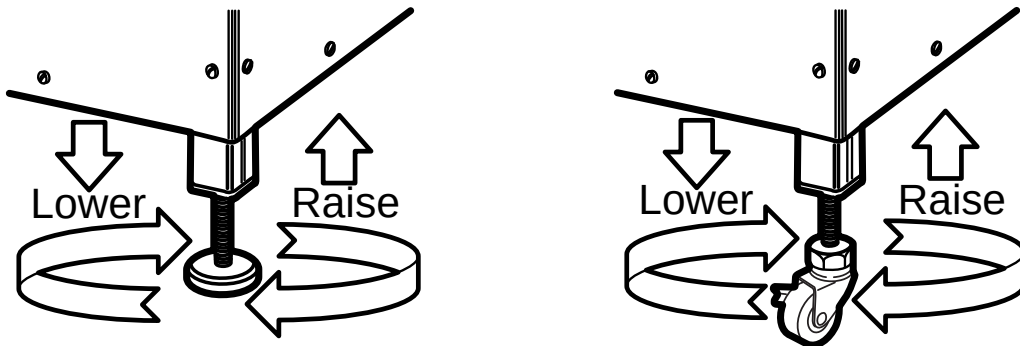


Adjustments

Shut off power and disconnect electrical connections before making any adjustments.

Machine Base Height and Leveling Adjustment

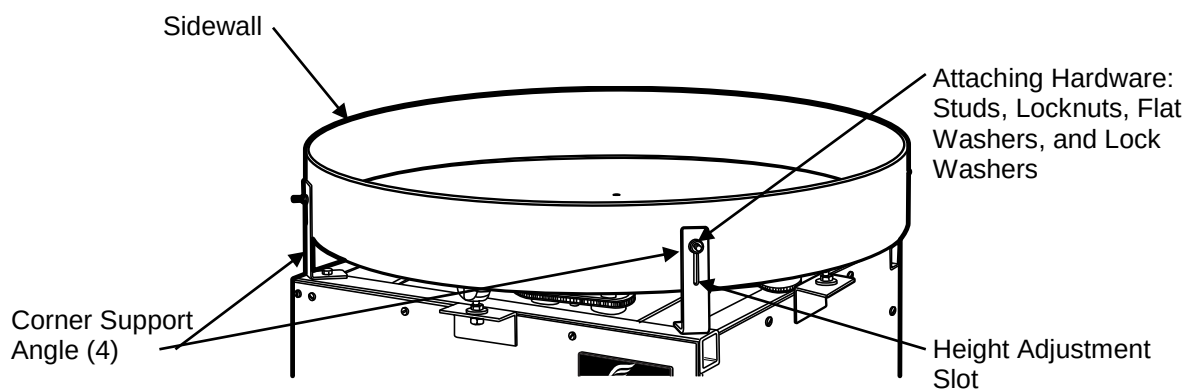
Each leg of the accumulation table is threaded to allow height adjustment. To adjust the leg height, twist each leg into or out of the accumulation table base to adjust the height up or down at each corner.



Note: When adjusting height to raise or lower the accumulation table base, place a level across flat continuous surfaces at the top of the turntable base and adjust the base to be level in both directions.

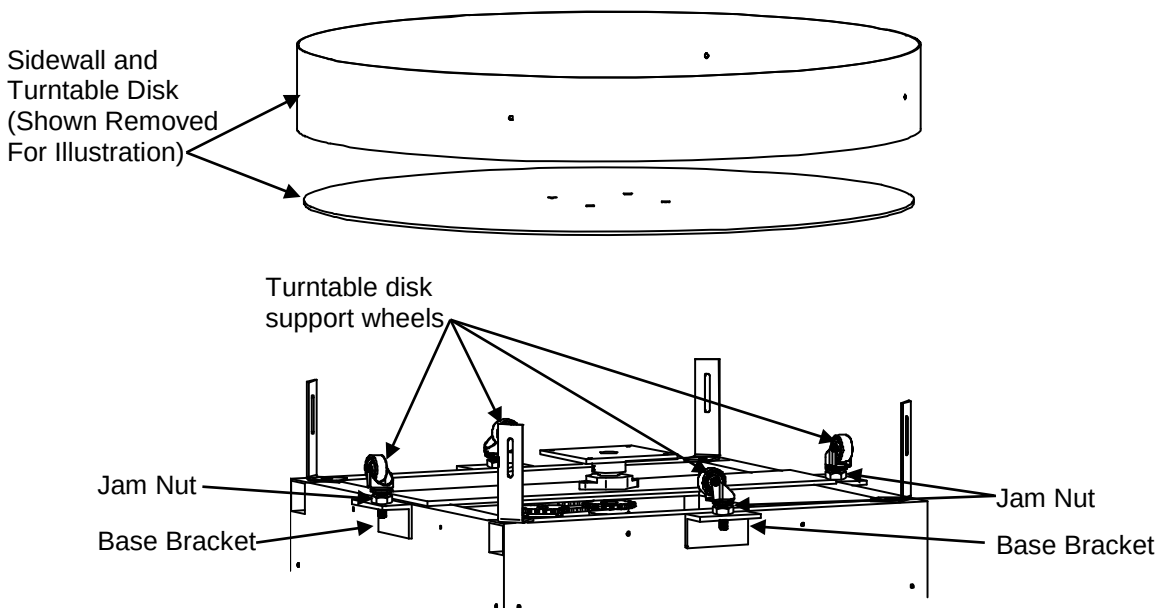
Sidewall Height Adjustment

Support angles at each corner of the accumulation table are slotted to allow height adjustment of the sidewall. To adjust the sidewall height, loosen the nuts and washers on threaded studs securing the sidewall to the support angle, adjust the height up or down at each corner, and then retighten at required height.



Turntable Leveling Adjustment

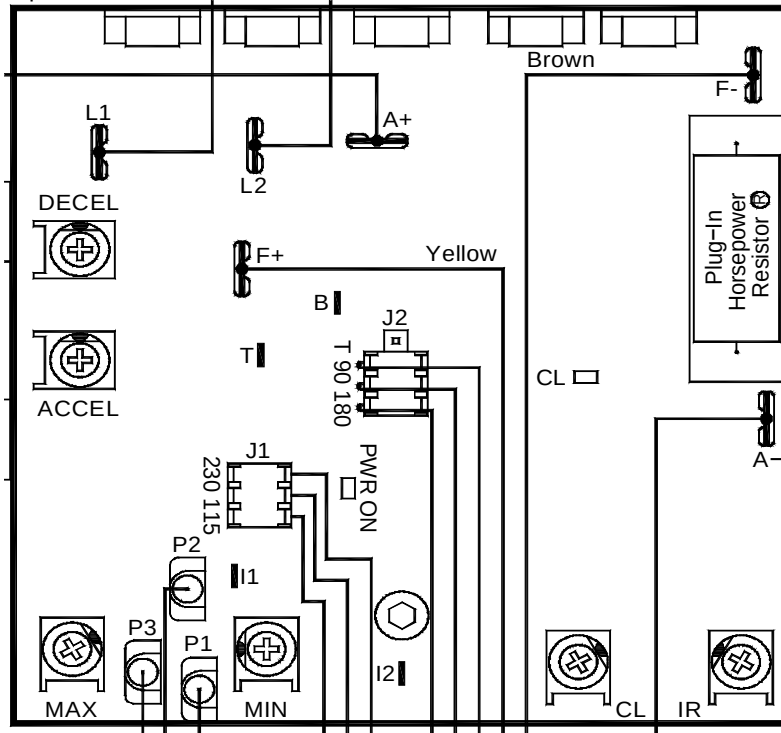
The four support wheels that are directly underneath and support the turntable disk are adjustable in height. After leveling the accumulation table base, check the turntable disk to make sure it is level in all directions to avoid any excessive vibration or wobble in operation. To adjust each wheel, first loosen the jam nut where it attaches to the base bracket. Turn the caster stud to adjust upward or downward as required, then re-tighten jam nut to secure height.



Advanced Speed and Control Function Adjustment

The following illustration shows the D.C. control board housed in the accumulation table control box. Characteristics of motor motion, such as maximum and minimum motor speed, motor acceleration and deceleration time, and current load and motor internal resistance compensation are fine-tuned by adjusting trim-potentiometers (trimpots) on this board.

Detail of KBMD-240D Multi-Drive™ Controller Board Showing Locations of Trim Pots

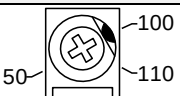
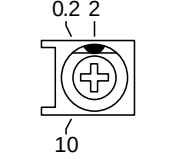
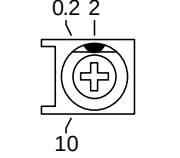
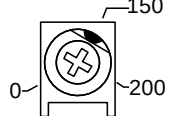


For more information refer to the *KBMD-240D Multi-Drive™ DC Motor Speed Control Installation and Operation Manual* (provided by the D.C. board manufacturer).

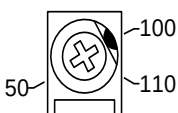
Trimpots on the internal multi-drive controller board have been factory adjusted to provide near-optimum motor regulation when using the control box speed and directional controls under normal operation. More specific information about the trimpots and adjustment is provided beginning on the following page. Refer to the figure above for location of trimpots.

Note: Do not attempt to change the setting of the trimpots unless absolutely necessary since they are factory adjusted to near optimum settings. Instructions in this section assume the proper Plug-In Horsepower Resistor® is installed for the particular motor and input voltage being used.

Typical factory-set trimpot settings are shown in the following table (expressed as value indicated on control board):

Appearance	Trimpot Descriptor	Value	Range
	MAX (maximum speed)	100%	50% - 110%
	MIN (minimum speed)	0%	0% - 30%
	ACCEL (acceleration start)	2	0.2 - 10
	DECEL (deceleration)	2	0.2 - 10
	CL (current limit/torque)	150%	0% - 200%
	IR (IR compensation)	3, 6	24, 48

Acceleration and Deceleration trimpots are labeled ACCEL and DECEL and are located near the left side of the speed control module. Maximum and Minimum trimpots are labeled MAX. and MIN. and are located near the bottom edge of the board on the left. Current Limiting and Internal Resistance trimpots are labeled CL and IR and are located near the bottom edge on the right. Refer to the figure on the previous page for location of trimpots.

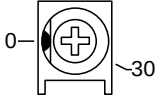


Maximum Speed Trimpot (MAX)

The MAX trimpot sets the maximum speed of the motor when the main speed dial potentiometer is set to fully clockwise. To decrease the maximum speed, adjust the MAX trimpot counterclockwise.

CAUTION! Do not attempt to adjust the maximum speed above the rated motor RPM since unstable motor operation may occur.

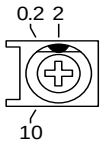
Note: For moderate changes to the maximum speed, there will be slight effect on the minimum speed setting when the minimum speed is set to zero. There may be significant variation in the minimum speed setting if the minimum speed is set higher than zero.



Minimum Speed Trimpot (MIN)

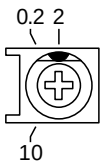
The MIN trimpot sets the minimum speed of the motor when the main speed dial potentiometer is set fully counterclockwise. The MIN trimpot has been factory set to 0% of the base motor speed. To increase the minimum speed, adjust the MIN trimpot clockwise. To decrease the minimum speed, adjust the MIN trimpot counterclockwise.

Note: **Readjusting the MIN trimpot will affect the maximum speed setting. Therefore it is necessary to readjust the MAX trimpot if the MIN trimpot has been adjusted. It may be necessary to repeat these adjustments until both the minimum and maximum speeds are set to the desired levels.**



Acceleration Trimpot (ACCEL)

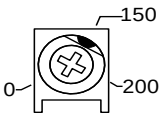
The ACCEL trimpot allows for a smooth start over an adjustable time period each time the AC power is applied or the main speed dial potentiometer is adjusted to a higher speed. The ACCEL trimpot has been factory set to two (2) seconds, which is the amount of time it will take for the motor to accelerate to full speed. To increase the acceleration time, adjust the ACCEL trimpot clockwise. To decrease acceleration time, adjust the ACCEL trimpot counterclockwise.



Deceleration Trimpot (DECEL)

The DECEL trimpot controls the amount of ramp-down time when the main speed dial potentiometer is adjusted to a lower speed. The DECEL trimpot has been factory set to two (2) seconds, which is the amount of time it will take for the motor to decelerate from full speed to zero speed. To increase the deceleration time, adjust the DECEL trimpot clockwise. To decrease the deceleration time, adjust the DECEL trimpot counterclockwise.

Note: **The deceleration time cannot be set to less than the natural coast time of the motor and actual load.**



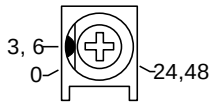
Current Limit Trimpot (CL)

The CL trimpot sets the current limit (overload), which limits the maximum current (torque) to the motor. The CL also limits the AC Line inrush current to a safe level during startup. The CL trimpot is factory set to 1.5 times the full load rating of the motor. To increase the current limit, adjust the CL trimpot clockwise (do not exceed 2 times the motor current rating (maximum clockwise position)). To decrease the current limit, adjust the CL trimpot counterclockwise.

To Recalibrate the Current Limit (CL) Trimpot

1. Disconnect the AC power and connect a DC ammeter in series with either motor armature lead. (Note: If only an AC ammeter is available, connect it in series with either AC line input lead.)

2. Set the Main Speed dial potentiometer to approximately 30% to 50%.
3. Set the CL trimpot fully clockwise. The CL LED will illuminate red.
4. Lock the motor shaft. Important: Be sure the CL trimpot is set fully clockwise and do not lock the motor shaft more than 2 to 3 seconds or motor damage may result.
5. Apply power and rotate the CL trimpot clockwise until the desired current reading is observed on the ammeter. Factory current limit setting is 1.5 times the full load rating of the motor with a DC ammeter connected in series with the motor armature. (If using an AC ammeter connected in the AC line input, the factory current limit setting will read 0.75 times the full load rating of the motor.) Do not exceed 2 times the motor current rating (maximum clockwise position).



IR Compensation Trimpot (IR)

When adjustment of the controller board is complete, it is necessary to reattach the controller box front cover. Slide the front cover in place to cover the controller box and re-use the two (2) #6-32 screws to fasten the front cover in place to the control box.

To Recalibrate the IR Compensation (IR) Trimpot

1. Set the IR trimpot to approximately 25% of rotation.
2. Run the motor with no load at approximately 1/3 speed and record the RPMs.
3. Run the motor with the maximum load and adjust the IR trimpot so that the motor speed under load is equal to the unloaded speed recorded in step 2 above.
4. Remove the load and recheck the RPMs
5. If the unloaded RPM has changed, repeat steps 2 through 4 for more precise regulation. The control is now compensated to provide minimal speed change with varying loads.

When adjustment of the controller board is complete, it is necessary to reattach the controller box front cover. Slide the front cover in place and cover the controller board and terminals and re-use the two (2) #6-32 screws to fasten the front cover in place to the control box.

For more detailed information refer to instructions in the previous pages or refer to the KBMD-230D™ Multi-Drive Controller Board Installation and Operation Manual, provided by the D.C. board manufacturer